



Society of Petroleum Engineers

SPE-229401-MS

Foaming Prediction AI System

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This document outlines the development of the Foaming Prediction AI System, an advanced digital solution designed to predict and prevent foaming incidents in AGR (Acid Gas Removal) facilities. The solution leverages machine-learning techniques to analyze operational parameters and detect early signs of potential foaming events.

The system utilizes a combination of operating data, engineering data, and machine-learning algorithms to identify patterns associated with foaming. Operating data are collected in real time to provide immediate feedback to plant operators. AI models are trained on historical operating data and engineering expertise to enhance predictive capability. The implementation process is streamlined and requires no modifications to existing hardware. Continuous data accumulation improves model accuracy and, in turn, stabilizes plant operations while increasing overall availability.

The introduction of the Foaming Prediction AI System, which has been successfully deployed at a Middle East facility, has reduced unscheduled turndowns and shutdowns due to the occurrence of foaming. The system's real-time analysis, which has enabled plant operators to take preemptive actions and has received positive feedback from on-site operators, reduces the risk of production loss. In detail, the implementation of the AI system has enabled the early detection and warning of foaming occurrence up to 30 minutes in advance, contributing to a 60-70% reduction in foaming. The AI system's continuous learning process has improved its predictive accuracy, leading to more stable plant operations. The effectiveness of the system in foam prediction demonstrated the effectiveness of the AI application in stabilizing plant operations and achieving higher Availability.

The Foaming Prediction AI System's novel approach, which integrates engineering expertise with machine-learning models for real-time analysis, represents a significant advance in proactive risk management. Its ability to learn from - and adapt to - new data delivers markedly better performance than that achievable with traditional monitoring methods.

Introduction

In recent years, the external business environment for plant owners in oil and gas industry has been subject to rapid and profound changes. Factors such as technological advancements, regulatory shifts, fluctuating market dynamics, and increasing environmental concerns necessitate innovative solutions that not only ensure operational safety but also enhance efficiency and sustainability. Thanks to the improvement of the speed of communication and processing data and the innovation of machine learning technology, plant owners are proceeding application of such state-of-the-art technologies to adapt and overcome those business environmental changes.

In those background, the key concept with digital technology for improving plant operation and maintenance is a concept of a Digital Twin. Digital twin use different kinds of models for a robust and flexible representation of the process. A key feature of digital twin is the adaptability and flexibility of the models to the real-time state of a plant through a cyber-physical system of sensors and data transfer protocols, automation, cloud computing and parallel computing [A]. Digital twin of chemical process can be especially called Process Digital Twin while a digital twin which utilizing 3D model is generally called 3D Digital Twin [B].

A process digital twin can be utilized for improving productivity, reducing operation and maintenance costs, and minimizing environmental impact by optimizing set points of process values or predicting anomaly incidents in plant operation, while machine learning is a key feature for the process digital twin in order to realize realized those achievements. We present technology and use case of early detection of foaming occurrence utilizing machine learning model with real operational data of AGR unit which is served as Foaming Prediction AI System.

Foaming in Acid Gas Removal Unit

The primary function of an AGR unit is to efficiently remove CO₂ and H₂S from sour feed gas using representatively an amine solution. However, a significant challenge faced in these units is the occurrence of foaming, which refers to the sudden and substantial generation of foam. Such foaming can severely disrupt the steady performance of acid gas absorption within the amine absorber column, making it a prevalent issue in the oil and gas sector.

The Impact of Foaming

Foaming not only impacts the operational efficiency of the AGR unit but also poses safety risks and environmental concerns. Negative impacts by foaming are [1]:

- Reduced efficiency and capacity: lower acid gas removal capacity, increase the circulation rate of amine
 - Increased operating cost : using antifoaming agents, frequent shutdowns, labor costs, maintenance costs by corrosion and erosion
 - Safety risks: pressure drop variations, equipment damage or vessel rupture
 - Product quality and purity: carry impurities (such as hydrocarbons, solids, or corrosion products) into the treated gas
 - Environmental impact: amine losses from the system, higher percentage of CO₂ and H₂S in the exported gas
 - Process control complexity: monitor and adjust process parameters, implementing effective foam control strategies
- Therefore, understanding the underlying causes of foaming and implementing effective predictive measures is essential for maintaining safe and reliable operations of AGR unit.

Causes of Foaming

One of the primary causes of foaming is the contamination of heavy hydrocarbons within the amine solution. This contamination can arise from various sources, including the feed gas composition, operational changes, and equipment wear. Additionally, the presence of impurities or surfactants can make foaming tendencies, further complicating the issue. Unfortunately, this contamination cannot be directly monitored by instruments, as it requires manual sampling of the amine solution to assess its condition. Consequently, identifying signs of foaming during real-time plant operations becomes exceedingly difficult.

Foam formation is typically characterized by the rapid expansion of gas bubbles within the liquid phase, leading to the accumulation of foam at the surface. This process can be influenced by several factors, including the viscosity of the amine solution, temperature fluctuations, and the presence of surfactants or impurities. As foam begins to form, it can create a barrier that hinders the effective absorption of acid gases, ultimately sacrificing the performance of the AGR unit.

The procedure to keep and maintain amine solution healthy is implemented as general operation and maintenance works. However, the factors of forming are very complex, and it is difficult to eliminate the cause perfectly all the time. Thus, foaming is a source of concern for operators, as they cannot predict when it will occur.

Detection Challenges

Detecting foam generation in real time presents a significant challenge for plant operators. Typically, foam generation can be detected through an increase in the differential pressure between the top and bottom beds of the column according to the DCS alert configured on the tag based on a specific differential pressure level, derived from past operational experiences. Once foam formation begins, it can expand dramatically within a matter of minutes, necessitating immediate action by operators. In such scenarios, field operators may be compelled to hastily dose anti-foaming agents into the amine system. Any delay in addressing foaming issues can escalate the risks of producing off-spec products, necessitating production downturns, or, in severe cases, triggering unscheduled shutdowns.

Given the critical nature of timely intervention, early detection of foaming incidents or its sign is crucial for operators to maintain stable plant operations. Proficient operators sometimes possess the ability to detect foaming tendencies, but this knowledge cannot be clearly articulated or implemented into standardized procedure.

Tan Kaiyun et al. explored fault detection and identification in AGR unit using deep autoencoders. Target type fault is foaming, damaged tray, fouling, solvent loss, and variation of feed gas composition. In the study, data made by process simulator ASPEN Hysys is utilized and it is succeeded to distinguish reasonably well between the different faults. [2]

Navin S et al. evaluated machine learning model using operational time-series data for early detection of foaming, while presenting correct identification of all severe foaming incidents by the site, and additional events bearing characteristic foaming signatures which were not logged by the site. [3]

The almost perfect number of the accuracy is the result of the simulation data source that runs in the ideal condition. However, industrial data tend to have an unknown operating factor, making it harder to reach higher accuracy [4]. In the fault detection application, deep learning is the best-performing algorithm due to its complex architecture and its ability to handle non-linearity and high data dimensions of AGRU in the real plant.[5]

Foaming Prediction AI System

Machine Learning Implementation

The Foaming Prediction AI System harnesses the power of machine learning to enhance predictive capabilities regarding foaming incidents. By utilizing a combination of operational data, engineering data,

and machine learning algorithms, the system identifies patterns associated with foaming occurrences. The AI models are trained on historical operational data, allowing for the generation of a foaming index that indicates the real-time risk of foaming.

The initial step in developing the system involves creating a comprehensive dataset that encompasses operational data during both foaming events and stable operational conditions. At this stage, operational data surrounding turnarounds and other incidents are excluded to ensure the dataset's integrity. Dataset comes from the tags of historian database through selection from the engineering viewpoints. Operational data of tags which are associated with causes and effects of foaming are selected among gas composition, flowrate, temperature, pressure, liquid level of AGR unit. Subsequently, the dataset is divided into training and validation subsets for applying supervised learning. The training data is used to train the AI model to recognize patterns and correlations, while the evaluation data is utilized to assess the model's predictive accuracy. The next phase involves generating a foaming index based on operational parameters related to foaming, selecting appropriate machine learning algorithms, fine-tuning parameters, and determining optimal learning conditions to enhance predictive accuracy. The collaboration of expertise for data science and engineering is essential for each steps.

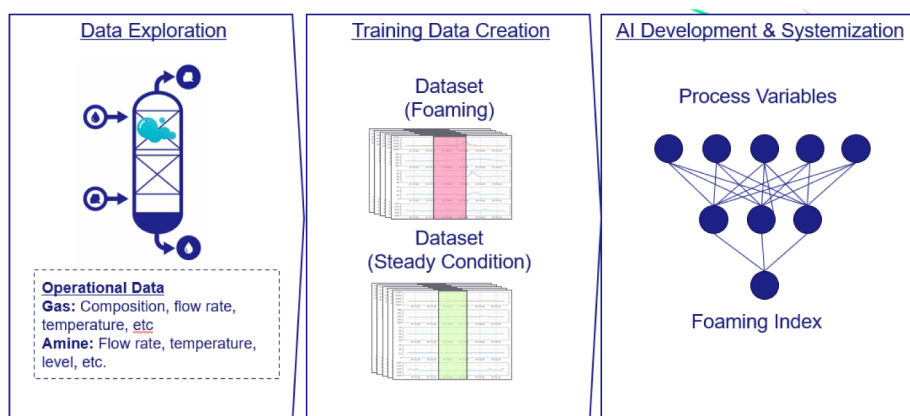


Figure 1—Steps of Machine Learning Implementation to Develop AI Model of Predicting Foaming

Furthermore, continuous accumulation of operational data contributes to the ongoing improvement of model accuracy, thereby stabilizing plant operations and enhancing overall availability. Since the conditions of feed gas or operational procedures may change over time, it is essential to continuously update the AI models to maintain their predictive accuracy. This adaptability ensures that the system remains effective under varying operational conditions and can respond to evolving challenges. Accumulated data is processed in the same manner as developing AI model periodically, and then the updated AI model is replaced in the system.

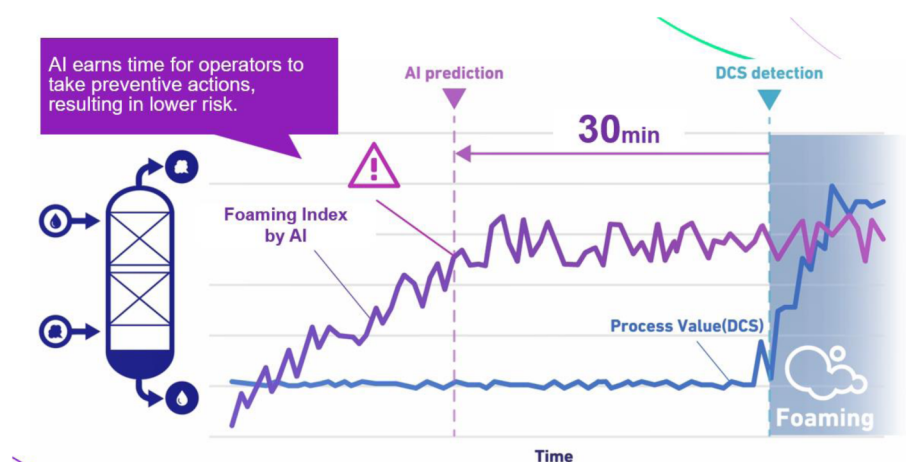


Figure 2—Value of Foaming Index Alert by AI Prediction Comparing to DCS Detection

System Features and User Interface

The interface of Foaming Prediction AI System is designed to be intuitive, featuring a process flow diagram of the AGR unit alongside the foaming index and alert notifications. This diagram illustrates the main equipment involved in the AGR unit and the process parameters that are closely linked to foaming occurrences. The foaming index is displayed in both current and trend values, allowing operators to monitor the risk of foaming in real time. Additionally, alerts provide timely notifications regarding potential foaming events, along with recommendations for corrective actions.

The system can function independently of the distributed control system, providing operators with a dedicated tool for monitoring foaming risks. This independence allows for greater flexibility and adaptability in managing foaming incidents, enabling operators to respond quickly and effectively. The system can be deployed on-premises or cloud basis as per the requirement of a company. A dedicated server for the system fetches operational data from the historian database, process those data for system use, and then calculate and respond the foaming index.

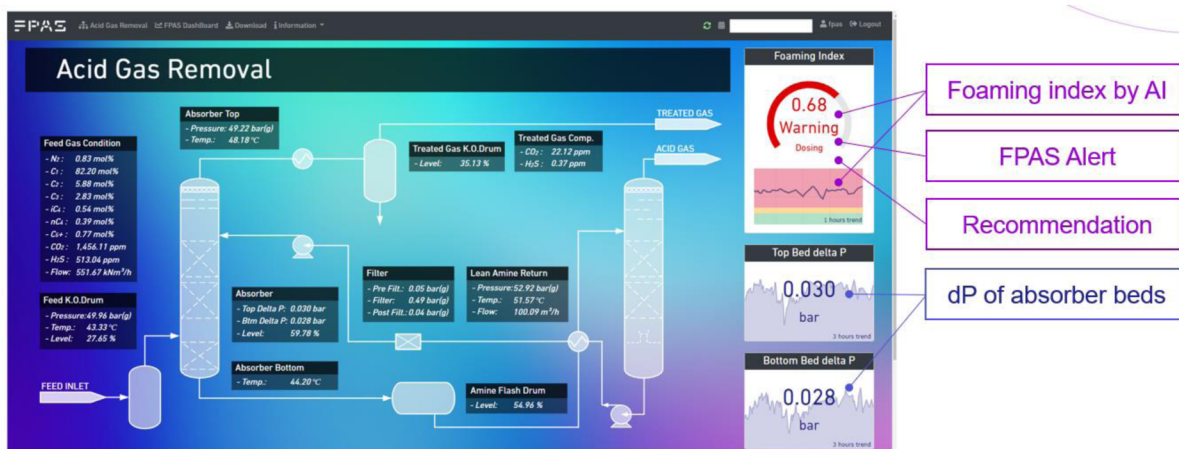


Figure 3—User interface of the Foaming Prediction AI System

Performance of the Foaming Prediction AI System

The Foaming Prediction AI system is deployed to a facility in the Middle East and has been evaluated. Accuracy of prediction is evaluated by two measurements: recall, the ratio of foaming successfully predicted, and precision, the ratio of true alerts by the system. Evaluation result is that recall is 80% and precision is 50%.

The score of recall intends that most of all foaming can be detected earlier than conventional method such as DCS alerts of differential pressure of the top and bottom beds of the column, while enabling proactive countermeasures against foaming. The time that it took to dose an anti-foam agent is measured by the duration of continuous differential pressure of the top and bottom beds of the column at a value higher than a certain threshold due to the occurrence of foaming and compared before and after the implementation of this AI system. It was confirmed that the duration of high differential pressure is reduced by 10%.

The score of precision intends that few false alerts still remain. However, those scores are acceptable since it is more important not to overlook the occurrence of foaming for stable operation. The balance of recall and precision, that is sensitivity of foaming alerts, can be adjustable based on the threshold of foaming index and operation policy of a company.

More specifically, the AI system facilitates the early detection and warning of foaming occurrences up to 30 minutes in advance, resulting in a remarkable reduction of foaming incidents by 60-70%.

During the evaluation period, the plant faced a feed gas composition that it had hardly experienced in the past. The AI model learns patterns at the time of foaming and stable operation from past operational data, so there was a temporary decrease in accuracy, but soon after that, it was confirmed that accuracy was restored by learning new data patterns with the automatic periodical learning function.

The successful implementation of the Foaming Prediction AI System at the facility has yielded significant benefits, including a reduction in unscheduled turndowns and a mitigation of shutdown risks associated with foaming incidents. The system's capability to conduct real-time analyses has empowered plant operators to take preemptive measures, thereby minimizing the risk of production loss.

Panel and field operators at the site are always worried about foaming that may occur at any time surrounded other works and troubles. Earlier detection of foaming is expected to reduce the mental and physical strain on those operators. It can also be expected that even inexperienced operators will be able to deal with foaming appropriately, rather than relying on the experience and intuition of proficient operators.

Future Development Directions

The development of the AI model for predicting foam generation has reached a stage of sufficient effectiveness for practical application at operational sites, fostering increased plant availability. However, to further enhance prediction accuracy, the digitalization of manual operations within the amine system and the collection of additional data regarding the amine solution are critical next steps.

As the amine solution undergoes partial or complete replacement, the operational preconditions will significantly change. This includes the cleaning or exchange of filters within the amine system, which enhances the purification capabilities of these filters. Converting information regarding these events into data and integrating it into AI model calculations is feasible during the training phase using historical data, yet it presents challenges during operational phases. Nonetheless, given the assumption that these factors influence the likelihood of foaming occurrences, it is essential to leverage this information effectively. Additionally, laboratory data regarding the amine solution can serve as a valuable resource for improving predictive accuracy. While it is crucial to acknowledge that laboratory data points differ significantly from time-series data such as flowrate or pressure, they provide direct insights into the state of the amine solution, making them indispensable for enhancing the AI model's performance as one of possible option.

Conclusions

The Foaming Prediction AI System represents a novel approach that integrates engineering expertise with advanced machine learning models for real-time analysis and proactive risk management. The AI model achieves enough high accuracy and duration for early detection of foaming occurrences even utilizing historical data for training and real operational data for calculating foaming index. Moreover, its ability to learn from and adapt to new data ensures continuous improved performance compared to traditional

monitoring methods. In addition to the performance of the AI model, usability as a system such as user-friendly interface is also key factor to mitigate actually risk of foaming since the intervention of the operators is essential. The successful deployment of this system not only enhances operational efficiency but also significantly contributes to the overall safety and reliability of acid gas removal processes, resulting in a reduction in unscheduled turndowns and a mitigation of shutdown risks associated with foaming incidents.

The development and implementation of the Foaming Prediction AI System exemplify the transformative potential of AI in the oil and gas industry. Process digital twin has its potential not only for foaming incidents, but also for optimizing set points of process values or predicting anomaly incidents in plant operation. As the sector continues to navigate complex challenges, the integration of advanced technologies will be essential for driving innovation, enhancing safety, and achieving long-term sustainability.

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